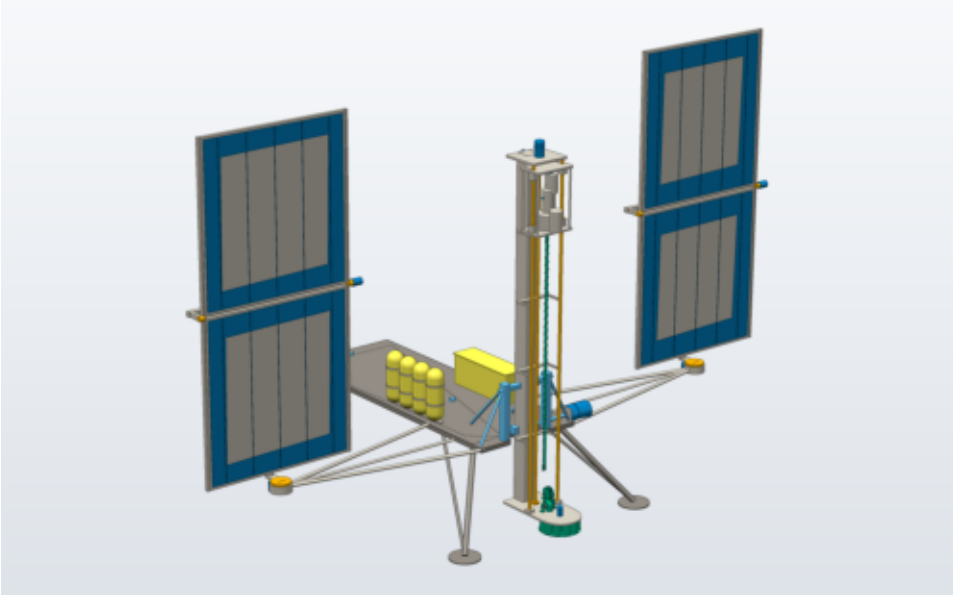
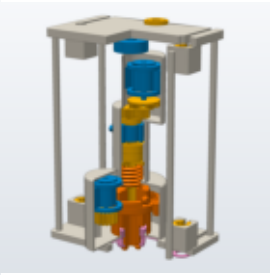
Autonomous Rotary-Percussive Bite Drill for the Moon and Mars

1 m ice-bearing samples from a 45 kg lander, with no drilling fluid: physics-based design, mission simulation and CAD

Jay Tilak P. · Model-based systems engineering in Python (build123d, numpy)

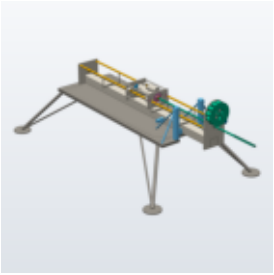




Drill head section



WC bit crown



Stowed for launch

Results vs the Icebreaker drill field test ("1-1-100-100")

Metric	Icebreaker	Moon	Mars	vs benchmark
Depth (m)	1.00	1.00	1.00	meets
Mission time (h)	1.00	0.96	0.76	meets
WOB set-point (N)	100	52	119	Moon meets, Mars above
Drilling power (W)	100	68	105	Moon meets, Mars above

Same order of magnitude as the closest field-tested prototype (Icebreaker) with no coefficient fitted to the benchmark. Mars WOB (119 N) is above Icebreaker's 100 N because the set-point is lander weight / 1.4 and Mars gravity is 2.3x the Moon's.

Key design decisions and the physics behind them

Percussion, not more push.
The lander can only press 73 N on the Moon before lifting off; rotary cutting of ice-cemented regolith needs more. A cam-and-spring hammer gives 2.25 J at 26 Hz (5.4 kN peak, stress-wave model).

Bite drilling, not continuous augering.
Without fluid, flights cannot lift cuttings at a safe speed (lift ratio 0.64 < 1 on the Moon). Flutes hold one 10 cm bite (choking at 12.2 cm); a passive brush strips them into that bite's own sealed cup on a carousel.

Carbide chisel bit, not PDC.
Two WC-10Co inserts, -10° rake, R1.0 hone sized by Hertz contact on a basalt clast (FOS 2.1); diamond would chip under impact and overheat dry.

Ice-preserving thermal control.
Bit temperature limit 164 K (Moon, sublimation) / 263 K (Mars, melt); peak simulated 120 / 205 K.

Deployable side-mounted mast.
80x80x3 6061-T6 Al spine chosen from 240 candidates; stowed 97 Hz (≥ 50 Hz launch), SMA hold-downs.

Tracking solar wings.
Sun ~1.5° above the lunar horizon: 2-axis tracking (azimuth + elevation) wings give 797 W (Moon) / 143 W (Mars).

Honest limitations / next steps

- Deployed mast mode 20-21 Hz sits only 2 % from the blow-rate band edge: a stiffer spine or tuned blow rate is the first change to test.
- Side mount: lander tips before it lifts; tilt margin 1.15 vs 1.4 for a centred drill.
- Rotating mast mass 25 kg drives a 108 N m deploy actuator.
- Brushing takes 87 s per bite (16 min per mission), the largest single time item after warm-up.
- Rock-cutting and heat-partition coefficients come from literature ranges; sensitivity sweeps are shown, but cryo-vacuum drilling tests are needed to calibrate them.

How it was built

One YAML parameter set (every input tagged as environment, design choice, coefficient or criterion, with sources) feeds a derivation ledger of 176 checked quantities, a 1 Hz mission simulation, and 286-part parametric CAD with STEP export. Change an input and every number, model and drawing on the site regenerates; automated tests guard the physics.

Interactive digital twin, derivation ledger, CAD (STEP/GLB) and drawings on the project site.